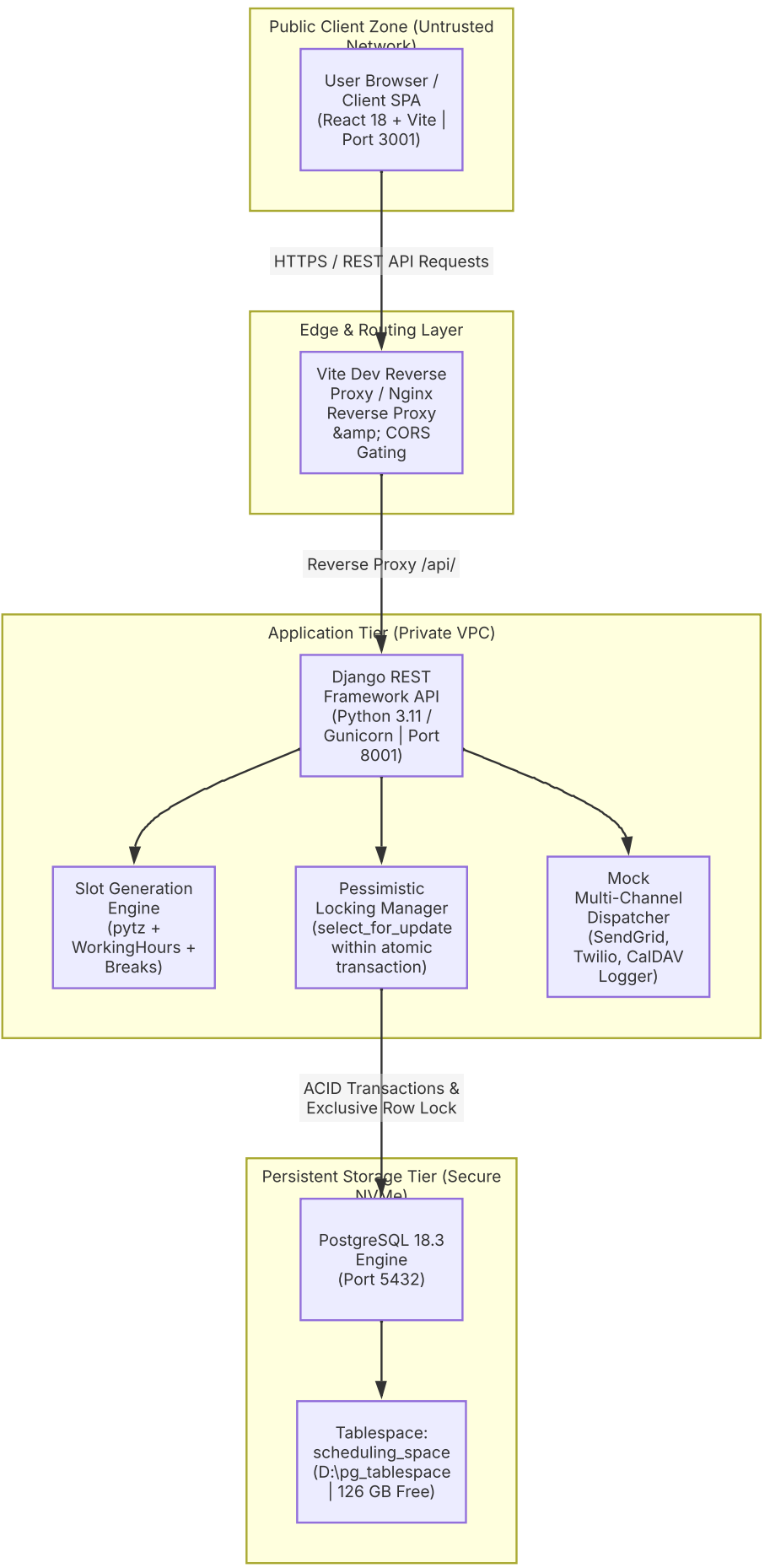
Public Client Zone (Untrusted Network)
User Browser / Client SPA (React 18 + Vite | Port 3001)
HTTPS / REST API Requests
Edge & Routing Layer
Vite Dev Reverse Proxy / Nginx Reverse Proxy & CORS Gating
Reverse Proxy /api/
Application Tier (Private VPC)
Django REST Framework API (Python 3.11 / Gunicorn | Port 8001)
Slot Generation Engine (pytz + WorkingHours + Breaks)
Pessimistic Locking Manager (select_for_update within atomic transaction)
Mock Multi-Channel Dispatcher (SendGrid, Twilio, CalDAV Logger)
ACID Transactions & Exclusive Row Lock
Persistent Storage Tier (Secure NVMe)
PostgreSQL 18.3 Engine (Port 5432)
Tablespace: scheduling_space (D:\pg_tablespace | 126 GB Free)